LAS 6292: Data Collection and Management

Emilio M. Bruna

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Introduction

This book is a manual for students in LAS 6292: Data Collection & Management. The course is designed for graduate students from any discipline – social sciences, humanities, biophysical sciences – and at all stages of their graduate program. It is an introduction to methods for collecting, organizing, managing, and visualizing both qualitative and quantitative data. Students will gain hands-on experience with best practices and tools.

Describe the different types of research data & the research data life-cycle

Explain the need for and benefits of data management and sharing

Describe and implement best practices for data collection, storage, management, and sharing

Find, download, and analyze publicly available data from repositories

Carry out simple and reproducible data corrections and dataset organization

Describe public policies and agency requirements for data management and sharing

Articulate the major legal & ethical issues regarding the data collection, use, and storage

Create and Implement Data Management Plans in funder-specific formats

Identify and properly use tools for more efficient and secure data collection in the field

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Citation

This guide is a Quarto Book hosted on the Bruna Lab's Github site. Please suggest edits to the text, missing resources, or make suggestions for improvement either by pull request or by posting an issue on the course book's repository. For more information and tutorials for doing so, see **?@sec-manual**.

Part I.

Course Topics

1. Naming Conventions

1.1. Introduction

The first (and easiest) way to get started organizing your data is to use a simple, clear, and consistent system for naming your files (i.e., a "naming convention"). Using a well-thought out naming convention has a number of important benefits:

- Files will be easier to find
- You won't have to open files to see what they are
- Files are easier to sort
- Files are easier to share with collaborators (and for collaborators to use)
- It helps prevent accidentally overwriting or deleting files

1.2. How to name your files

Your file names should:

1. **Tell you about the contents of the file and allow you to uniquely identify it.** For instance (compare `'data 1998'` vs `'survey data'` vs `'survey data 1998'` vs `'small mammal survey data 1998'`). You can use things like the an acronym for the project, abbreviation for the study, location where collected, the name of the investigator, the year or month when collected, the type of data type, etc.

2. **Be as short as possible.** Many database systems (like MS One Drive) have limits on file name length. Use a max of 25 characters, and aim for less than that.

3. **Avoid using special characters and spaces** Don't use characters like `$ % ^ & # | :`. This is really important because software, automated data processing tools, web browsers, and computer operating systems use spaces and special characters for dividing up (parsing) strings of text, as well as other processes.

4. **Start with letters, never numbers.** Using numbers at the start of file names can lead to problems when they are read into stats software and other programs. Instead of `'2020_survey_responses'` use `'survey_responses_2020'`

5. **Help you quickly sort files chronologically or numerically**. Dates are very useful in file names because they can help identify the most recent files easily. You can *not* rely on the file date associated with the file on the computer! However, it is really important to use a clear format for dates, especially if collaborating internationally. I learned the hard way that not everyone reads `'census_data_3-4-2018'` the same way I do, so I started using `'census_data_4march2018'`. However, even this isn't ideal because of

how it sorts things in a folder. Now I now use this format: `'file_name_YYYYMMDD'` (`'census_data_20180403'`). You can also use `file_name_YYYY-MM-DD`. This is based on the international format for dates and times (ISO 8601). In addition to the benefit of sorting more easily, you'll never have to use the the dreaded `'_____.final.docx'` and `'_____.actual_final.docx'` in your file names ever again…you'll always know which is the last one. If the files don't require a date, but you want to maintain a sequence, use leading zeros to maintain sort order. `7` and `701` don't sort the way you expect them to, so use `007` and `701`.

6. **Use a consistent method of dealing with spaces and letter case.** . For a the spreadsheet with responses to a survey by 5 different people, don't use `'survey responses Florida'`. Instead use (i) `'survey_responses_florida'` or (ii) `'SurveyResponsesFlorida'`. Try to avoid mixing the two (e.g., `'survey_responses_Florida'`). By the way, (i) is known as `pothole` and (ii) is known as `camel`. Personally, I prefer pothole name is better because of it is easier to read and you never to remember what words get capitalized (because none of the do). Use underscore (`_`) or dashes (`-`) to separate meaningful parts of file names.

Once you have a system, write it down, print it out, and put it up somewhere you can easily refer to it. Remember -whenever possible make the names simple, informative, and unique. If you collected survey data and behavioral data in 2017, don't call them both `data_2017` even if they are in different folders. Call one `'behavior_obs_2017'` and the other `'survey_responses_2017'`).

Simple names also help you avoid problems with "system and software portability" - you might be working on a Mac but your collaborators are working on Linux or Windows, and the software made for each can read file names in different ways. File names that are simple, lower-case, and have no special characters are less software- and platform-dependent.

1.3. Tools & Resources

1.3.1. Bulk Renaming of Files

- Adobe Bridge
- Bulk Rename (Windows)
- Rename Master (Windows)
- File Renamer Basic (Windows, Free and Paid)
- Ant Renamer
- Renamer 5 (macOS)
- Renamer Lite
- PSRenamer
- Batch Renaming of Files in Windows: Post 1, Post 2
- Batch Renaming of Files in MacOS: Post 1, Post 2

1.3.2. Best Practices - Other Institutions

- Smithsonian Library

- Michigan Library
- Stanford Library
- Smithsonian

2. File Organization

2.1. File Organization - Electronic

There aren't really "rules" about how to organize your files, but I think the best way to organize is to think in terms of "projects". Of course, how you define a "project" is sometimes tough…is the PhD the 'Project' or is each chapter of the dissertation a 'Project'? You'll have to give this some thought and think of a system that works for you. That said…

I like 'Project' based organization for two main reasons:

1. **First**, it pushes you to think in terms of outcomes: 'projects' lead to 'products' (e.g., paper, report, grant proposal). This helps you think more clearly about what data need to be collected and the analyses that need to be done. If the data aren't for a new project, and it's not clear to you if or how the data fit into a pre-existing project, then are you sure you need to spend time, effort, and money collecting them?
2. **Second**, project-based organization focuses on the individual steps and components that take a project from start-to-finish: planning your data collection, data gathering and clean-up, analysis, presentation, and data archiving.

2.2. Example: Project Based Organization

A project-based organization scheme for a master's thesis might look something like this:

1. Top Folder: emb_masters_thesis
2. Sub-folder 1: `Chapter1_lit_review`
3. Nested Folders and Subfolders for:
    - Plans for Data Collection & Management
        - DMP
        - plans for storage and backup
    - data
        - raw data
        - clean data
        - metadata
    - analyses
        - intermediate outputs from analyses
        - final outputs from analyses
    - presentation (e.g., `masters_ch1_text` or `ch1_slides`)

- data archiving

You can have one of these for each thesis chapter. Note that some (DMP, plans for storage and backup) might be better placed one folder up in the `emb_master_thesis` folder

But what if I will be using the same data for multiple chapters or papers? This is a great questions, and there are several options. One would be to include both chapters in the same folder, each with its own subfolder (manuscript 1, manuscript 2). The other would be to have each as a separate project, and pull the data in from the online archive where you deposited the data (or folder where the data are stored). As you can probably guess, I think #2 is better because I think of each paper as a different project.

2.3. File Organization - Physical

Whenever possible I try to organize physical records and archives in the same way as digital ones: in a single box, file folder, cabinet, etc.** try to set it up so that you go to one place for everything related to a project. Of course, you can't always do that because specimens go to museums or stay in an archive…but you can keep the information on how to *find* the originals there. Don't forget to include copies of the survey, data sheets, etc., as well as maps or other such resources.

Correspondence is a bit more complicated. You can keep notes - electronic or paper - in the relevant folder. But one thing that has gotten complicated is email correspondence regarding projects, manuscripts, etc. Printing seems like a huge waste of paper, and yet if critical it might be worth doing. Other wise keep an email folder for each project as well - with the same name as the electronic version so you can easily find email correspondence related to each project.

2.4. Documentation

Write it down. Prepare a file documenting your data organization structure so you can share it with team members (or look at it years later when looking for something). These can be `'README.txt'` files as needed for each folder. The `'README.txt'` file might include such things as explanations of naming conventions and how the structure of the directory relates to the structure of the project.

There are some excellent tools for organizing projects. Use appropriate tools, such as version control tools or electronic lab notebooks, to keep track of the history of the data files (more on this later in the semester).

3. Data Organization in Spreadsheets

3.1. Make your data `tidy`

- Spreadsheets should be a rectangle, with only rows and columns.
- Each column is a different variable (a thing you are measuring, like 'weight' or 'temperature').
- One row per observation. Each cell has only one value.
- Column headers: Use short meaningful column names with no spaces or special characters. Don't start column names with numbers. Record units in column headers.
- Don't enter the same data on multiple spreadsheets: Use one for each category of data to avoid duplicated data and to simplify corrections (e.g., taxonomy).
- Never put multiple tables in a single spreadsheet.
- Avoid spreading data across multiple sheets
- *Collecting* data in tidy format makes it easier to *enter* data in tidy format.

3.2. Use consistent names, abbreviations/codes, and capitalization.

- Write dates as YYYYMMDD. Better still have separate columns for Year, Month, and Day.
- Excel is unable to parse dates from before 1899-12-31. Be careful if your data include a mix of dates before and after this date, then you'll have mixed data types in one column. [1]
- Record zeros with a numeral (`0`), not a blank cells. For missing data use an appropriate null value indicator (e.g., `NA`).
- Don't use formatting to convey information or to make your spreadsheet look pretty.
- *Remember that data format and excel defaults can vary by region.* For example, depending on the part of the world where a user is based, the default value for the decimal and thousands operator could be a `,` (comma) or a `.` (period); some regions use mm-dd for dates while others use dd-mm.

[1]The reason dates in Excel are so weird is that it is *accounting software*. It counts the days from a default of December 31, 1899, and thus stores July 2, 2014 as the serial number 41822. This is so one can can easily calculate "days from a given date" for accounting purposes (like invoicing) by adding "date+XX days".

3.3. DO NOT EDIT OR CORRECT RAW DATA FILES!

- Once you are done with data entry, save your file in 'read only' format and make *all* corrections using scripting.
- Do *not* edit raw data after you have entered it in your spreadsheet!*

3.4. Readings, Tools, & Resources

1. Broman, K. W., & Woo, K. H. (2018). Data organization in spreadsheets. The American Statistician, 72(1), 2-10. [read online]

2. Data Validation in Google Sheets: blog post and video tutorial. A pdf version is available for download here.

3. Why not bypass spreadsheets like Excel and use a csv editor like [Comma Chameleon][https://comma-chameleon.io/] instead? CC and other csv editors allow you to enter data in the same way - into cells, by adding and removing rows - and then export your file. But that's about it, which means you can't do many of the things (e.g., calculations, color in cells) that cause problems down the road.

4. More advanced users comfortable with R can also look into Data Curator, with which you can create and edit tabular data from scratch or from a template, open Microsoft Excel and CSV files, and automatically correct common problems found in these and other file types.

5. DataONE Community Engagement & Outreach Working Group (2017) "Data Quality Control and Assurance". Accessed through the Data Management Skillbuilding Hub at https://dataoneorg.github.io/Education/lessons/05_qaqc/index on Aug 31, 2020

6. DataONE Community Engagement & Outreach Working Group (2017) "Data Entry and Manipulation". Accessed through the Data Management Skillbuilding Hub at https://dataoneorg.github.io/Education/lessons/04_entry/index on Aug 31, 2020

7. Chris Prener, Trevor Burrows (Eds.). Data Carpentry: Data Organization in Spreadsheets for Social Scientists.

8. Peter R. Hoyt, Christie Bahlai, Tracy K. Teal (Eds.), Erin Alison Becker, Aleksandra Pawlik, Peter Hoyt, Francois Michonneau, Christie Bahlai, Toby Reiter, et al. (2019, July 5). datacarpentry/spreadsheet-ecology-lesson: Data Carpentry: Data Organization in Spreadsheets for Ecologists, June 2019 (Version v2019.06.2). Zenodo. http://doi.org/10.5281/zenodo.3269869

Part II.

In-Class Activities

4. Course Introduction: In-Class Work

4.0.1. This week's assignment is to briefly answer the following questions. *If you respond to the Google forms survey there is no need to submit anything via Canvas*

This survey has two goals:

1) To learn a bit more about your interests and experience so that I can tailor the modify the course content to match the cohort's interests.
2) To gather some information the data set(s) you might use for the Data Cleanup Project: one of the goals this semester is to take a "messy" data set, clean it, organize it, and – assuming you have permission and if appropriate at this time – archive it in a repository. This assignment will introduce me to your data set so we can evaluate if it is suitable for the assignment (i.e., is it messy enough, is it big enough?). The data can be from previous research, even if it is unpublished or not part of a thesis.

It is important to remember that the 'data set' can be more than one file. For instance, your data set might be four files (or collections of files): (1) A file with the responses to surveys of community members, (2) Census data downloaded from a government website, (3) The transcripts of interviews with key informants, and (4) A collection of photographs for ethnographic research.

Please also remember that you may not have a data set with which to work. That's fine! There are plenty of (useful) data sets out there that are messy and in need of organization. I'll help you find one that matches your interests.

4.1. Survey Questions

1. UFID Number
2. Last Name (as it appears on Canvas)
3. Do you have a name or nickname by which you prefer to be called?

4. Canvas has an option for students to list their preferred pronouns. If your preferred pronouns aren't available as an option on Canvas, please feel free to tell me what they are here.

5. What degree are you pursuing?

   - MA/MS
   - PhD
   - Other

6. What is your department or degree program?

7. Please provide a brief summary - 1 paragraph max - of your research interests.

8. Do you have any questions or concerns about the class you'd like to share with me? Perfectly acceptable answers include: "I'm worried about the workload", "I've never taken a bio class", "I'm worried we might have to move on-line in a few weeks because of covid,"I'm worried classes are inperson instead of on-line because of covid", and "None".

Potential Data Set for Course Project

9. Do you have Messy Data you would like to clean up and organize as your course project? This can be data from your current or prior research, including side projects, Master's thesis, undergraduate thesis, etc. *If you don't have a data set of your own, THAT'S NO PROBLEM - I will help you find one.*

   - Yes
   - No
   - Maybe

10. What research question(s) are being addressed with these data?

11. What is the status of this data set?Leave only the most relevant answer.*

    - I have finished collecting the data (Skip to question 13)
    - I have started data collection but will *not* be done before the semester ends (Skip to question 13)
    - I have started data collection and *will* be done before the semester ends (Skip to question 13)
    - Other (please elaborate) (Skip to question 12)

4.1.1. Data Set Status = Other

12. If you answered "other" to the previous question, briefly elaborate here.

4.1.2. Tell me about the data set(s)

13. What are they and what kind of information do they contain?
14. What tools and techniques did you use to collect it?
15. What format are they in (e.g., sheets of paper, notebooks, spreadsheet, text files, photographs)
16. How much data is it? (estimated number of files, surveys, photographs, etc).
17. Who collected the data?
    - I did / I am / I will
    - A 3rd party (e.g., my advisor, another student, a collaborator)
    - It it is previously published or public data Other:
18. Are the data from research conducted with human subjects?
    - Yes
    - No
19. Are any collaborators currently working with the data?
    - Yes
    - No
20. Will anyone (e.g., collaborators, advisor) be working with the data in the future?
    - Yes
    - No
    - Don't Know
21. What instruments did you use for COLLECTING and RECORDING raw data (check all that apply)
    - paper survey or data recording forms
    - online surveys
    - notebooks/forms in which personal observations, notes, or responses to questions were recorded
    - audio recordings
    - data from automated data loggers
    - photographs
    - video
    - medical instruments (EKG, eye trackers; please elaborate below)
    - tablets/smartphone
    - Other (elaborate in "Data Collection" section below)
22. What were you gathering on or about? Check all that apply.
    - field-based experiments, observations, or data collection with live animals or plants

- museum collections: herbarium specimens, animals (including tissue or blood), fossils
- plant or non-human animal tissue or blood (not from a museum specimen).
- human subjects - behavior, observations, experiments (education, psych, linguistics, etc)
- human subjects - tissue, blood, or similar human subjects - records (medical, educational)
- museum collections - human remains
- museum collections - archeological (pottery, middens mounds)
- museum collections - anthropological
- museum collections - art (painting, sculpture)
- Other (elaborate in "Data Collection" section below)

4.1.3. Data collection Instruments & Subjects

23. If you answered "other" to either of the 2 previous questions, please elaborate on the data collection instruments or subjects.

4.1.4. Instruments

24. What instruments and software do you use for PROCESSING and ORGANIZING data? Check all that apply

    - MS Word
    - Google Docs
    - Other word processor
    - MS Excel
    - Google Sheets
    - Other spreadsheet software
    - Scrivener
    - OSF
    - Evernote
    - Flickr
    - Text editor
    - Other (elaborate below)

25. If you answered "other", please list the other tools you use for PROCESSING and ORGANIZING your research data
26. If appropriate for your research domain, do you use any of the following to analyze your data?
27. If you answered "other", please list the other tools you use for ANALYZING your research data
28. Are you an R user? NOTE: YOU ARE NOT EXPECTED TO HAVE ANY PROGRAMMING EXPERIENCE TO TAKE THIS CLASS.

    - I have never used R

- Yes - a novice R user
- Yes - an intermediate R user Yes - an advanced R user

29. Will you be gathering data this summer or fall?

- Yes
- No
- Not Sure

30. Is there anything you'd like to add? Are there particular topics you'd like to see covered or tools you'd like to learn?

4.2. Grading Rubric:

1. Survey completed with thorough answers: 50

2. Most questions answered completely; some require instructor follow-up: 40

3. Many questions missing answers or answers are cursory: 30

4. Instructor follow-up required for survey submission: 20

Part III.

Assignments

5. Assignments Overview

Grades in the course will be based on the following assignments:

5.1. Weekly In-class activities

- **Overview**: Most of the in-class assignments involve hands-on practice with data collection or manipulation. In some weeks, however, assignment will be the submission of questions for group discussion or brief reflection on the issues from the readings. Most in-class assignments are designed to be completed during the class session, but to ensure students master the concepts rather than rush through them *they can be submitted anytime until 9 am the following Friday.*

5.2. Reproducible Data Organization Project

- **Overview:** This project is an opportunity to put some of the lessons learned into practice with a data set of your own. Your assignment is to **(1)** clean and organize a 'messy' data set and prepare metadata describing the resulting 'clean' data. The complete project requires the submission of these three items via the course Canvas website:

  (1) R code that imports, cleans, and organizes, and saves 'messy' data

  (2) The resulting corrected and organized data in an appropriate format

  (3) Metadata describing the corrected data set

5.3. Thesis Research DMP

- **Overview:** The Data Management Plan (DMP) is a critical document describing the data to be collected for a research project, how it will be stored and managed, and the investigator with primary responsibility for its management. Many funding agencies, including NSF and NIH, now require a DMP with all grant applications. Each student will prepare a Data Management Plan (DMP) for their thesis research.

References

Part IV.

Appendix 1: Useful Resources

6. Useful resources for Data Collection & Managment

6.1. R Programming

6.1.1. Essential

1. Hadley Wickham wrote a book on using the tidyverse and the online version is FREE. This is a phenomenal resource on using R to import, tidy, and visualize data.
2. Posit Cheat Sheets: help with commands for using the different `tidyverse` packages, RStudio shortcuts and tricks, help with R commands, and more. You definitely want the ones for Data Import, Work with Strings, Factors, Data Transformation, and Base R.
3. Where and How to ask for help
    - Hadley Wickham's advice on how to write a good reproducible example for getting help with R
    - how to post good questions on StackOverflow
    - The UF R-users listserv is *very* user friendly and a great place to post requests for help.

6.1.2. Tutorials and Books

1. R Essential Training: Wrangling and Visualizing Data
2. Software Carpentry: Using RStudio for Project Organization & Management
3. Swirl
4. R Bootcamp
5. Kieran Healy's *Data Visualization: a practical introduction* is my favorite introductory (yet super-comprehensive) book on data visualization with R. If you scroll down to the bottom of the page you can download the datasets and code used to make the figures in the book, which makes life much easier.
6. So. Many. Resources.
7. ROpenSci: tools for accessing, manipulating, and visualizing open data

8. How to clean messy data in R
9. The Ultimate Guide to Data Cleaning
10. Learning R
    Swirl

7. Specific Problems in Data Cleaning and Managemnt

1. Handling dates and times in R
2. Text Mining: `tidytext` package
3. Working with Qualtrics survey data with the `qualtRics` package
4. Optical Character Recognition (OCR): extract text from images: `tesseract` package
5. Extract text & metadata from pdf files: `pdftools` package
6. Image processing in R: the `magick` package

7.0.1. Advanced R Packages

1. `DataCurator` package: 'a simple desktop data editor to help describe, validate and share usable open data'.
2. RegExr: online tool to learn, build, & test Regular Expressions (RegEx / RegExp)
3. janitor (cleanup of file names, etc.)
4. `knitr` overview: reproducible documents with R
5. qualtRics
   ## Discipline-specific Resources
6. `historydata` package: Sample data sets for historians learning R. They include population, institutional, religious, military, and prosopographical data suitable for mapping, quantitative analysis, and network analysis.
7. *The Programming Historian* Website: wide range of topics, from text analysis to OpenRefine

7.1. Slide Presentations in R

1. Make slide presentations with R

7.2. Data Archives

Qualitative Data Repository

7.3. Text Extraction and Organization

Plan for extraction and organization

7.4. Form Design

Best Practice for Form Design

7.5. Data Security

UF Office of Information Security and Compliance
Cyber Safeguards for UF
UF IRB
UF Data Classification Policy
UF Office of Information Security and Compliance

7.6. Platforms for Organization & Collaboration

Open Science Framework

8. Slide decks

Here are the slide decks:

- Methods — Slides
- History — Slides